

Lab 12 - Prompt and Specification Pack

Cloud-Edge Architecture Design and Measured Trade-offs Generative AI for Image and Video Creation
(TGS-2020505925) | Version v11.0 | 6 September 2026

This pack is the architecture decision record and the review prompts that keep a cloud-versus-edge argument evidence-based. Every figure it produces carries its own provenance: MEASURED, MODELLED or ILLUSTRATIVE.

Currency note. Model identifiers and API parameters quoted anywhere in this pack were verified against the official vendor documentation on 6 September 2026. Model IDs change; re-verify before relying on one. Identifiers known to be deprecated are marked as such rather than quietly removed, so you can recognise them in older material.

Architecture decision record (ADR) template

```
ADR-<n>: split point for the NorthQuay generative visual pipeline
STATUS: proposed | accepted | superseded

CONTEXT
  41 stores, <uplink> Mbit/s per store, camera feed classified <class>.
  Latency budget: <ms> for shelf alerts; <s> for asset generation.

CONSTRAINT GATES (applied first, not weighted)
  privacy: <which splits are eliminated and by which rule>
  uplink : <which splits are arithmetically infeasible>

MEASURED
  per-stage latency (median / p95 ms) on <CPU>, OpenCV <version>, <n> frames
  end-to-end median <ms> against a budget of <ms>

MODELLED
  generation latency proportional to S (image) and S x F (video); S=<n>, F=<n>
  INT8 parameter memory = 8/32 = 0.25 x FP32
  low-rank adapter trainable params r(d+m) = <n> against d*m = <n>

ILLUSTRATIVE
  monthly transferred data <GB> at <rate>/GB = <cost> (classroom unit rates, not a quotation)
  energy per sample = overhead x energy-per-FLOP x FLOPs-per-sample

DECISION
  Split <X>. Runner-up split <Y>. Decided by: <the single number>.

CONSEQUENCES
  <what becomes easy, what becomes hard, what must be built>

RESIDUAL RISK
  1. <risk> -> closed by <action>
  2. ...
```

Protocol selection table

LINK	PAYLOAD CLASS	LATENCY CLASS	PROTOCOL	REASON
camera -> edge node	live video	real-time	RTSP	standard camera transport, low setup cost
camera -> browser preview	live video	interactive	WebRTC	sub-second, NAT traversal built in
edge -> cloud telemetry	small events	seconds	MQTT	pub/sub, tiny header; configure retries for
constrained sensor -> edge	tiny events	seconds	CoAP	a poor link UDP-based, for very low-power devices
app -> model endpoint	request/reply	seconds	HTTPS/REST	request/reply; caching depends on endpoint; verify support
service -> service	request/reply	sub-second	gRPC	binary, streaming, typed contracts
render -> asset store	large blobs	batch	HTTPS	object storage, resumable upload

Every row must state a payload class and a latency class. A protocol chosen without both is a preference, not a decision.

Split-point comparison prompt for the review meeting

For each of the four split points, give me exactly these six numbers:

1. bytes per second per camera at the split;
2. Mbit/s per store, and Gbit/s across 41 stores;
3. feasible on the stated uplink? yes/no;
4. measured end-to-end latency of everything left on the edge side;
5. what personal data crosses the split, if any;
6. illustrative monthly transferred data cost.

Then tell me which single number eliminates each rejected option.

Do not present a preference before presenting the six numbers.

Evidence-provenance rule (put this at the top of every architecture deck)

Every figure in this document is labelled:

- MEASURED - timed or counted on stated hardware, with n and the percentile.
- MODELLED - derived from a documented relationship, with the relationship shown.
- ILLUSTRATIVE - built on assumed unit rates; not a quotation and not a forecast.

An unlabelled figure is a defect and will be sent back.

Edge-fit checklist

- [] Weights quantised to 8-bit where the accuracy loss is measured, not assumed (parameter memory scales by $b/32$, so 8-bit is $0.25 \times \text{FP32}$).
- [] Sensitive activations kept at higher precision for numerical stability.
- [] Task specialisation via low-rank adapters: $r(d+m)$ trainable parameters instead of $d*m$, so the base weights are not duplicated per task.
- [] Sampler steps reduced by distillation, with the quality delta measured, not assumed.
- [] Latent-space modelling used at high output resolution to cut memory traffic.
- [] Structured pruning considered where the kernels support it.
- [] Every one of the above reports a before/after quality number, not just a speed-up.

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